TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Hartmut Eigenbrod	Email	Hartmut.eigenbrod@ipa.fraunhofer.de
Institute Name	Fraunhofer Institute for Manufacturing Engineering and Automation IPA			
Address	Nobelstr. 12, 70569 Stuttgart, Germany			

2. Technology Overview

Title of Proposed Technology	**Process Anomaly Detection and Analysis using Sensor and Text Data (PANDA)**			
Korea's 12 National Strategic Technology Fields	AI (√), Advanced Robotics & Manufacturing (√)			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		Mechanical・Materials	
	Medium Category	Small Category	Medium Category	Small Category
	R&D / Engineering Services	Product Quality Management Technologies	Robotics/ Automation Machinery	Measurement/Sensor Technologies for Automation
Concept of Proposed Technology	The proposed technology combines sensor-driven process monitoring with advanced AI analysis of free-text anomaly reports from production environments. Core functions include automated detection and classification of process anomalies, integration of structured process data and unstructured text, as well as interactive user interfaces. Potential applications are in industrial quality control, rapid root-cause analysis, and efficient process optimization in manufacturing.			
Rationale for Joint R&D Project	Efficient management of process anomalies in manufacturing requires expertise in both AI-based text analysis and process monitoring. Collaboration between Fraunhofer IPA and Korean companies leverages strong industrial backgrounds and complementary know-how, enabling validation in real-world settings and ensuring scalable, practical solutions for Korean companies.			
Key Words	(1) AI-driven Anomaly Detection, (2) Sensor and Text Data Fusion, (3) Industrial Process Optimization, (4) Large Language Models, (5) User Interaction Interface			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6-7)
R&D Objectives	**Initial situation:** The production of high-quality products requires integrated, continuous process monitoring to ensure product quality. This enables companies to detect anomalies early in production, providing a basis for detailed analysis, evaluation, and resolution. Efficient anomaly management is a key lever for improving economic efficiency. Depending on the specific production task, implementation typically includes: 1. A combination of process-integrated data acquisition and (manual) production assessment: Systems are installed in production to continuously monitor processes using sensors (see Figure 1, left). This includes documentation and evaluation of process anomalies. Due to the high variability of anomalies, documentation in industrial practice is often partly done in free-text form. 2. AI-supported systems for process monitoring and anomaly detection: To identify complex relationships in production data, process monitoring and anomaly detection systems are used. Due to the high dimensionality of the data, machine learning methods are often employed (see Figure 1, right). Detected anomalies are displayed to the machine operator. **Deficit:** Free-text fields contain valuable information about specific anomalies and errors in production. Currently, they are mainly used for immediate resolution of anomalies. Additionally, manual searches and analyses of free-text reports are performed for specific questions, usually by process or quality engineers. However, the analysis is very time-consuming due to the unstructured nature of the texts, and results cannot be automatically reused. In the IT and AI fields, powerful systems for automated text understanding are available, especially for the analysis and evaluation of free-texts. However, there is currently no practical link or propagation of data from free-texts (see point 1 above) with statistically operating automated process monitoring (see point 2 above). As a result, significant potential in the data for rapid and targeted resolution of process anomalies remains unused. **Overall R&D Objectives:** The goal of the project is to analyze and assess the potential for efficient resolution or prevention of process anomalies, based on automated integration of data from two sources: (1a) process data from process monitoring or anomaly detection systems, and (1b) free-text documentation of anomalies by machine operators. This is illustrated by the bold information pathway in the center of Figure 1. The project explores the expected improvement potential for process monitoring first in a feasibility study, then in a full implementation. In particular, the performance of task-adapted large language models (LLMs), which form the technology backbone of well-known tools such as ChatGPT, is evaluated. Fraunhofer IPA has already carried out relevant preliminary work on solution components.			

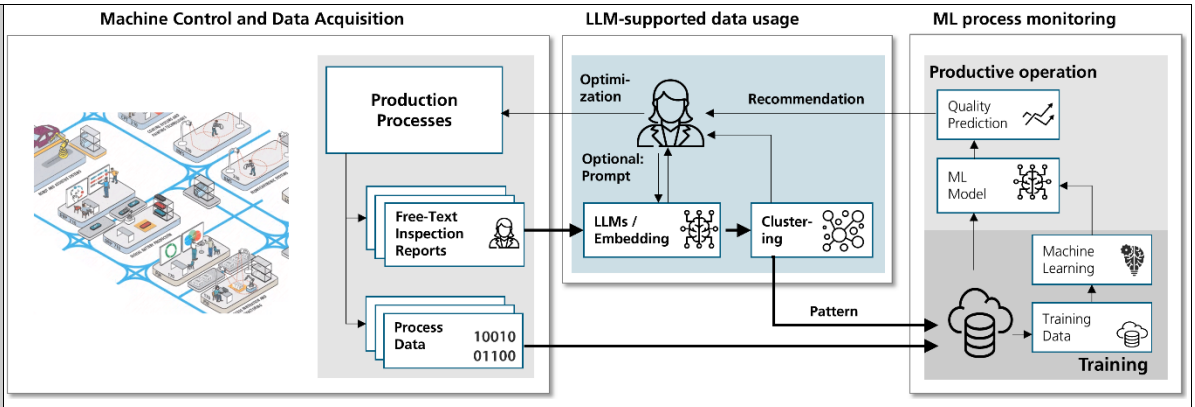

Figure 1: LLM-based use of free-text as a link btw. production and ML-assisted process monitoring

Specific R&D Objectives: Inspection Report Systems

State of the Art: In production processes, it is essential for continuous quality improvement that error descriptions and unexpected behaviors are documented and recorded. Furthermore, it is crucial during ongoing production to document deviations in both processes and products. In manufacturing companies, this documentation is generally performed as a combination of manual and automated methods.

The manual process is based on the creation of inspection reports after individual production steps, performed by responsible staff such as machine operators, quality engineers, etc. The specific content and extent of documentation vary greatly across industries and companies. The same applies to IT implementation. Widely used and relatively easy to set up are systems for entering free text, allowing identified errors or observed deviations to be described. This enables precise descriptions that support a thorough understanding during further error analysis. However, this form of documentation is challenging for recurring errors, beginning with the reliable retrieval of similar or related inspection reports. Joint evaluation of such cases is also labor-intensive. In addition to free text fields, there are often further structured input fields allowing the selection from pre-defined categories or the entry of numeric data.

Manually entered data is often merged with data from automated information and data acquisition systems, such as higher-level Manufacturing Execution Systems (MES) or automated quality monitoring systems.

In problem-solving teams or committees, it is crucial to identify root causes of errors quickly and unambiguously. However, poor quality free-text entries can negatively affect the discoverability of relevant information and subsequent analysis.

Current Limitations: At present, there are no easy-to-use ML models for industrial manufacturing that automatically evaluate free-text fields in inspection reports and utilize these results in existing ML-based process monitoring systems.

Status at Fraunhofer IPA: Experience with free-text in the context of inspection reports exists; for further details, see the status of Fraunhofer IPA in the area of LLMs.

Project Objective for this Section: The project's objective is to enable the automatic extraction and efficient use of valuable information from free-text inspection reports, integrating this data with existing process monitoring systems to support faster and more reliable root cause analysis and problem resolution.

Specific R&D Objectives: Large Language Models

State of the Art: The state of the art for Large Language Models (LLMs) has advanced remarkably in recent years, especially due to innovative approaches such as OpenAI's GPT models. These models are based on transformer architecture and are capable of processing large amounts of text data and generating human-like language. They are used in a variety of fields, including text generation, machine translation, content creation, and customer service.

Sources highlight that generative AI, including LLMs, has the potential to fundamentally change the manufacturing landscape. Through automation of routine tasks and decision support, LLMs can improve efficiency in production environments. For example, they can assist in creating technical documentation, analyzing production data, and optimizing workflows. Integration into digital assistant systems is also promising, making it easier for employees to access information.

LLMs are advanced AI models developed to generate human-like text. They have revolutionized many areas such as content creation, customer service, and software development. Furthermore, they play a central role in translation, summarization, and question-answering systems. LLMs differ in their architectures, training methods, and training data. There are three main categories of LLMs: basic LLMs are general, pre-trained models. Fine-tuned LLMs are further trained for specific tasks or domains. Mixture-of-Experts (MoE) models consist of a group of LLMs, each handling different parts of the input.

Current Limitations: Despite these advances, there are challenges to overcome. These include managing bias in training data, ensuring model interpretability, and protecting personal data. Companies must also develop guidelines for responsible use of AI systems. Future developments could focus on these topics to fully realize the potential of LLMs in industry while complying with ethical and legal standards.

Status at Fraunhofer IPA: Prototype implementation of LLMs for free-text evaluation of inspection reports exists (see references). Further development is needed, especially regarding how the extracted information can be used in subsequent monitoring systems.

Project Objective for This Section: The project aims to further adapt and refine state-of-the-art large language models for industrial free-text analysis, ensuring robust, explainable, and responsible AI that delivers actionable insights for process monitoring and improvement.

Specific R&D Objectives: ML-based Systems for Process Monitoring

State of the Art: The state of the art in ML-based process monitoring systems in industrial manufacturing has made significant progress in recent years. Machine learning enables precise monitoring and analysis of production processes by processing large volumes of data in real time. These systems identify anomalies, optimize processes, and improve quality assurance through predictive analytics. Algorithms such as decision trees, neural networks, and support vector machines have proven effective. Modern approaches like edge computing enable fast, on-site data processing, thus increasing efficiency and minimizing downtime, which leads to higher productivity.

Current Limitations: These monitoring systems are currently not designed to use free-text from inspection reports as input data.

Status at Fraunhofer IPA: Fraunhofer IPA has previously implemented various ML-based monitoring methods in different industries and production technologies such as plastic injection molding,

ultrasonic bonding, assembly processes, etc. So far, no free-text information (neither raw data nor summaries) has been used in these applications.

Project Objective for This Section: The objective is to extend ML-based process monitoring systems with the capability to seamlessly incorporate and utilize insights derived from free-text inspection reports, enhancing overall accuracy, responsiveness, and value of industrial anomaly detection and quality assurance.

R&D Target Specifications

The target performance specifications for this R&D project:

- Improvement in anomaly detection accuracy: Increase anomaly detection rate by at least 15% compared to current sensor-only systems.
- Reduction in manual analysis time: Decrease operator time spent on evaluating inspection reports by 30%.
- Integration efficiency: Achieve seamless data integration with at least 95% successful data linkage between free-text reports and process data.
- User acceptance: At least 80% positive feedback from operators in usability tests.

R&D Plan

This section outlines the main work packages, detailing key tasks and deliverables to ensure systematic progress and effective collaboration throughout the project.

WP1: Project Scoping (Months 1–3)	
Objectives	Establish overall project framework, align expectations, and define initial requirements.
Tasks	- Kick-off meeting; define communication and reporting structure. - Identify stakeholders and assign roles. - Review existing documentation and prior solutions. - Joint workshops to identify relevant use cases. - Develop project objectives, success criteria, risk management, and IP strategy. - Prepare initial data management and security plan.
Deliverables	Project charter, stakeholder map, consolidated use case list, project management handbook, risk register.

WP2: Pre-study (Months 4–6)	
Objectives	Analyze technical feasibility and collect pilot data to inform main R&D activities.
Tasks	- Survey production lines and select pilot sites. - Assess availability and quality of historic sensor and text data. - Implement pilot data collection protocol if needed. - Data exploration: profiling, annotation needs, gap analysis. - Benchmark available ML/LLM technologies. - Identify integration and infrastructure challenges. - Review regulatory requirements.
Deliverables	Feasibility study, pilot data report, initial technology assessment,

	recommendations for main R&D phase.

WP3: Data Acquisition and Preprocessing (Months 7–12)	
Objectives	Prepare robust, high-quality datasets for model development using historic data where possible.
Tasks	- Identify and gather historic sensor and inspection report datasets. - Evaluate and select relevant datasets in collaboration with partners. - Annotate and validate datasets (anomaly labels, metadata, etc.). - Address data quality issues: noise filtering, missing data imputation, normalization. - Ensure compliance with data privacy and security standards. - Set up database and version control for datasets.
Deliverables	Curated, annotated datasets; data pipeline documentation; data quality assessment report.

WP4: Model Development (ML & LLM) (Months 12–27)	
Objectives	Develop advanced models for anomaly detection, free-text analysis, and data fusion.
Tasks	- Develop baseline ML models for anomaly detection. - Fine-tune/adapt LLMs for industrial free-text (German, Korean, English). - Explore prompt engineering and domain adaptation for LLMs. - Implement data fusion strategies. - Develop explainability methods. - Iterative training and validation using annotated data. - Document models, training, and evaluation metrics.
Deliverables	Trained ML and LLM models, model performance reports, fusion algorithms, explainability tools.

WP5: System Integration and Prototyping (Months 18–30)	
Objectives	Integrate developed models into a prototype system and connect with production environments.
Tasks	- Design and implement system architecture. - Develop user interfaces for operators and engineers. - Integrate model outputs with MES/SCADA systems. - Ensure real-time or near real-time data flow and inference. - Establish secure data exchange interfaces. - Functional/interface testing in lab. - Document integration and user guidance.
Deliverables	Prototype system, integration documentation, user manual.

WP6: Pilot Deployment and Evaluation (Months 25–33)	
Objectives	Validate system performance and collect user feedback in real production settings.
Tasks	- Deploy prototype at pilot sites in Germany and Korea. - Monitor operation and collect performance/user data. - Conduct user acceptance testing. - Optimize models and system components. - Quantitative evaluation against KPIs. - Assess scalability and robustness.
Deliverables	Pilot evaluation report, user feedback analysis, recommendations for improvement.

WP7: Dissemination and Exploitation (Months 30–36)	
Objectives	Secure long-term impact through dissemination, training, and exploitation planning.
Tasks	- Organize workshops, webinars, and demos. - Prepare training materials and conduct training. - Publish results in journals and magazines. - Develop exploitation plans for both markets. - Finalize IP agreements and business models. - Prepare final project report and policy recommendations.
Deliverables	Dissemination materials, training documentation, exploitation plan, final project report.

Milestones

- **M1 (Month 3):** Project charter, use case definition, management structures established
- **M2 (Month 6):** Pre-study complete, pilot sites and data sources validated
- **M3 (Month 12):** Curated, annotated datasets and pipelines ready
- **M4 (Month 27):** ML and LLM models developed and validated; fusion strategies operational
- **M5 (Month 30):** Integrated prototype system ready for pilot deployment
- **M6 (Month 33):** Successful pilot deployment and evaluation at industrial sites
- **M7 (Month 36):** Final dissemination, training, and exploitation activities completed

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Technical Contributions:** • Lead the research and development of machine learning (ML) and large language model (LLM) algorithms for anomaly detection and text analysis. • Provide expertise in data fusion methodologies. • Design and implement prototype systems, including integration with industrial IT environments. **Research Scope:** • Develop, train, and validate AI models using industrial sensor and text data. • Evaluate system performance and user acceptance in pilot deployments. **Resources available at Fraunhofer IPA:** • Access to state-of-the-art research infrastructure and computational resources. • Experienced researchers, data scientists, and software engineers. • Annotation tools, model development environments, and documentation templates. **Specific Tasks:** • Coordinate project management and reporting. • Support data acquisition and preprocessing, including data quality assessment. • Lead model development, training, and testing. • Organize dissemination activities, user training, and scientific publication. Fraunhofer IPA will ensure high-quality R&D outcomes, knowledge transfer, and effective collaboration with industrial partners throughout the project.
Expected R&R of Korean SME Partner	**Technical Contributions:** • Provide (anonymized) data from relevant production lines, sensor data, and inspection reports. • Contribute domain knowledge on manufacturing processes and operational requirements. • Support adaptation and deployment of AI solutions within real industrial environments. **Development Tasks:** • Assist in data acquisition, annotation, and validation using in-house resources. • Participate in requirements engineering and system customization. • Collaborate in the development and integration of software modules into existing IT systems (e.g., MES/SCADA). **Testing/Validation Activities:** • Host pilot deployments and facilitate real-world system testing. • Provide feedback on system usability, performance, and reliability from an operator's perspective. • Contribute to quantitative and qualitative evaluation of project outcomes. **Resource Commitments:** • Dedicate technical staff for project activities and on-site support. • Ensure secure data sharing and access for research purposes. • Allocate infrastructure for prototype installation and testing. **Collaboration Efforts:** • Engage in regular project meetings and joint workshops. • Work closely with Fraunhofer IPA for knowledge exchange and technology transfer. • Support dissemination and exploitation activities, including industry outreach. The Korean SME will play a crucial role in ensuring industrial relevance, validating solutions in real settings, and facilitating successful market transfer.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to deliver an integrated AI-based system for efficient detection and analysis of process anomalies, leveraging both sensor data and free-text inspection reports. The project aims to achieve significant technological advancements, such as improved anomaly detection accuracy and reduced manual analysis effort, with successful demonstration in real industrial environments. The resulting solutions will be transferable and scalable, allowing adaptation across diverse manufacturing sectors. By fostering close collaboration between Fraunhofer IPA and SMEs, the project will accelerate technology adoption and innovation. There is strong commercialization potential through the development of new software modules, system upgrades, and consulting services tailored for industrial AI applications. Overall, the project will have a broader industrial and economic impact by increasing productivity, reducing errors and waste, supporting digital transformation, and enhancing the global competitiveness of German and Korean manufacturing industries.

6. References

(1) Grigorjan, Arthur; Amirkhanian Namagerdi, Hayk; Sielaff, Lennard; Enhancing Error Report Analysis in Industry: Evaluating Large Language Models and Traditional NLP for Text Clustering, Automation, Robotics & Communications for Industry 4.0/5.0; Heft 2024, S. 264-269; ISSN 2938-4796

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(3) Buchner, Christoph; Seidler, Christian T.; Huber, Marco F.; Eigenbrod, Hartmut; Ribbeck, Hans-Georg von; Schlicht, Franz (2024): Machine learning-based shear force quality prediction of ultrasonic wire bonds. In: Int J Adv Manuf Technol 133 (11-12), S. 5657–5672.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization